

Handout 2.18: Return Math

Name: _____ Date: _____ Partner (Part C): _____

The formula

Percent return = (ending value minus starting value) divided by starting value, times 100.

As a picture: a fraction with “ending minus starting” on top and “starting” on the bottom, then “times 100” beside it. (The slide shows it drawn large with **STARTING** in bold.)

The bottom of the fraction is always the **starting** value. If you divide by the ending value, you get the wrong answer every time.

A loss gets a minus sign. A gain gets a plus sign.

Worked example. An investor started with \$1,000 in 1870 and has \$850 now.

- Ending minus starting: 850 minus 1,000 = minus 150
- Divided by starting: minus 150 divided by 1,000 = minus 0.15
- Times 100: minus 15 percent

The investor’s return is minus 15 percent.

Part A: Practice (alone)

Show the subtraction line and the division line for each one. Every price is from the board.

#	Starting value	Ending value	Ending minus starting	Divided by starting	Percent return
1	\$1,000 (a ledger in 1870)	\$1,250			____ percent
2	\$1,000 (a ledger in 1870)	\$850			____ percent
3	\$50 (one share of RAIL in 1870)	\$30 (1873, the Panic)			____ percent
4	\$60 (one share of OIL in 1911)	\$70 (1928)			____ percent

Grade 6 support rows (round numbers):

#	Starting value	Ending value	Ending minus starting	Divided by starting	Percent return
5	\$100	\$120			____ percent
6	\$100	\$75			____ percent

7. The trick question. GAS was \$40 in 1928 (the peak) and \$20 in 1932 (the bottom).
- The drop, as a percent of the peak: (20 minus 40) divided by 40, times 100 = ____ percent
 - The climb back, as a percent of the bottom: (40 minus 20) divided by 20, times 100 = ____ percent
 - Why are the two percents different when the dollar change (\$20) is the same both ways? Answer: _____

Part B: My own return (alone, from your own ledger)

My starting value, 1870: \$1,000

My total right now, at Round 8 (1932) prices: \$_____

Ending minus starting: _____

Divided by starting: _____

Times 100: my percent return so far is _____ percent (circle: gain / loss)

Check it on the calculator: _____ Same? yes / no

The company that hurt me most from the peak to the bottom. From your price table: the Round 6 price (1928, the peak) and the Round 8 price (1932, the bottom).

Ticker: ____ 1928 price: \$____ 1932 price: \$____

(1932 minus 1928) divided by 1928, times 100 = ____ percent

Part C: A safe ledger, a one-company ledger and a mixed ledger over the same rounds (pairs)

- Safe:** all \$1,000 in the bank.
- One company:** 20 shares of RAIL at \$50 (\$1,000), nothing else.
- Mixed:** 5 RAIL at \$50 (\$250), 25 OIL at \$10 (\$250), 12 GAS at \$20 (\$240), and \$260 in the bank.

Ledger	Value at the peak (1928)	Return at the peak	Value in the crash (1929)	Return in the crash	Value at the bottom	Return today
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					(1932, today)	
Safe	\$1,195	plus 19.5 percent (done for you)	\$1,231	plus 23.1 percent (done for you)	\$1,268	plus 26.8 percent (done for you)
One company	\$1,400	_____ percent	\$900	_____ percent	\$200	_____ percent
Mixed	\$2,891	_____ percent	\$2,179	_____ percent	\$1,245	_____ percent

The Safe row is worked for you as a model. Compute the other six. Round to one decimal place.

8. Which ledger looked smartest (a) at the peak, (b) in the crash, and (c) today?
(a) _____ (b) _____ (c) _____
9. Which ledger would you rather have held for these 62 years? Circle one. Then say what it is better for (money for college in ten years, or rent next month) and give a number.

Safe / One company / Mixed. Because:

More room: _____

10. Grade 8 stretch. The one-company ledger's fall from the peak to the bottom, as a percent of the peak: $(200 \text{ minus } 1,400) \text{ divided by } 1,400, \text{ times } 100 = \text{_____ percent}$. To get back from \$200 to \$1,400 it must climb: $(1,400 \text{ minus } 200) \text{ divided by } 200, \text{ times } 100 = \text{_____ percent}$. Explain in one sentence why a ledger that fell 86 percent has to climb 600 percent to get back.

My sentence: _____